Kernix 0.7.3 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

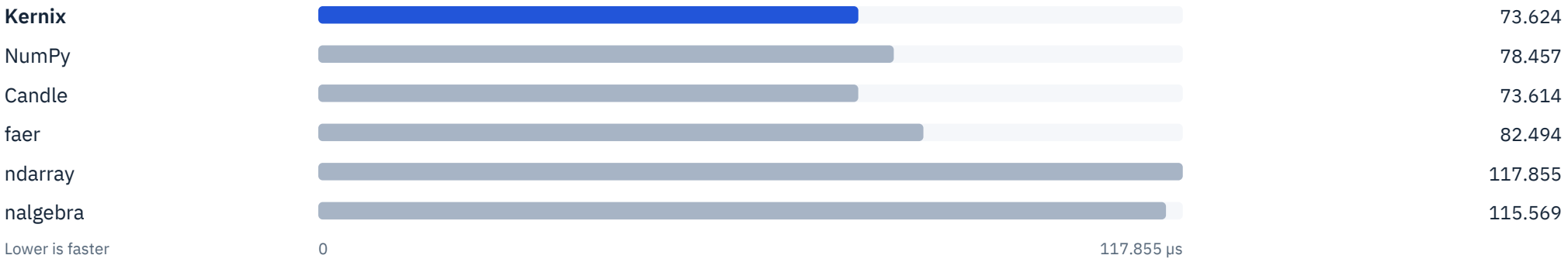
19 workloads

80 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

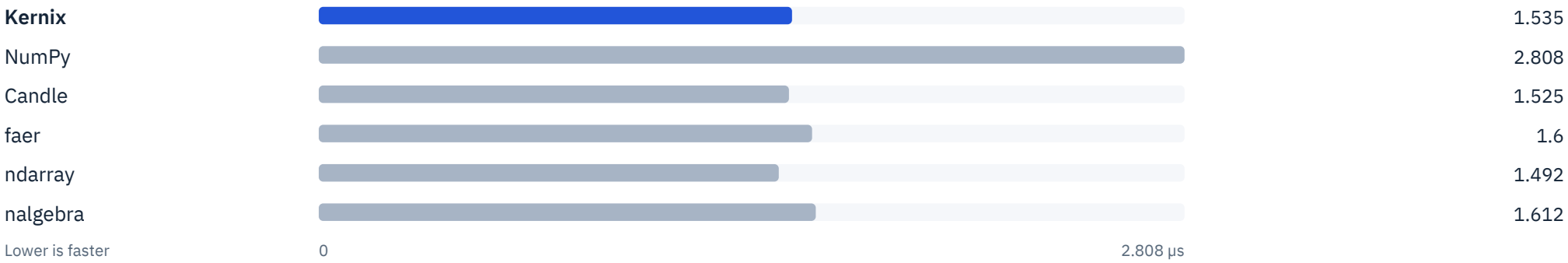
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

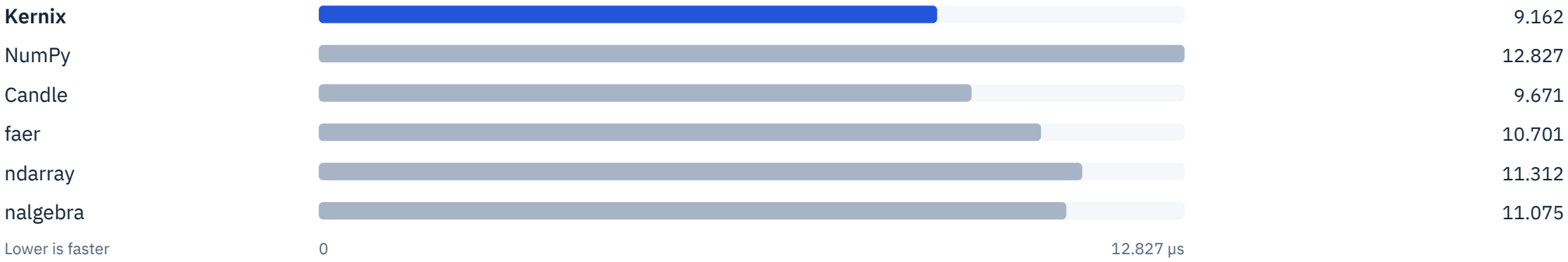
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



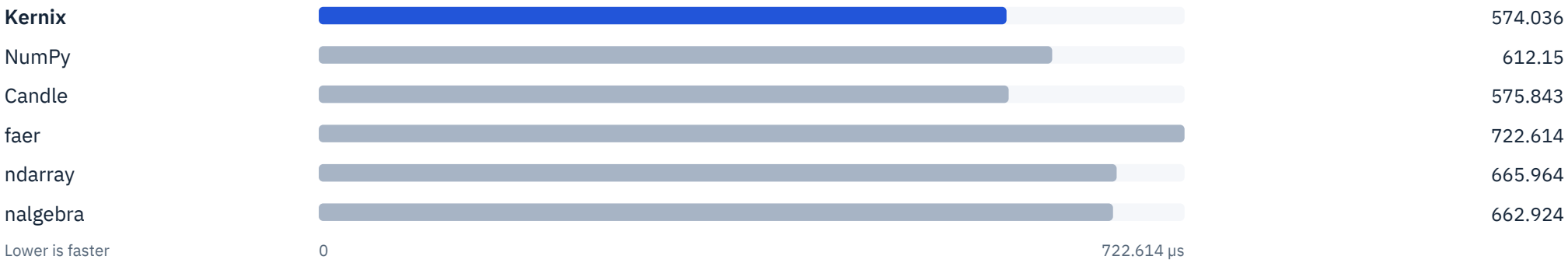
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

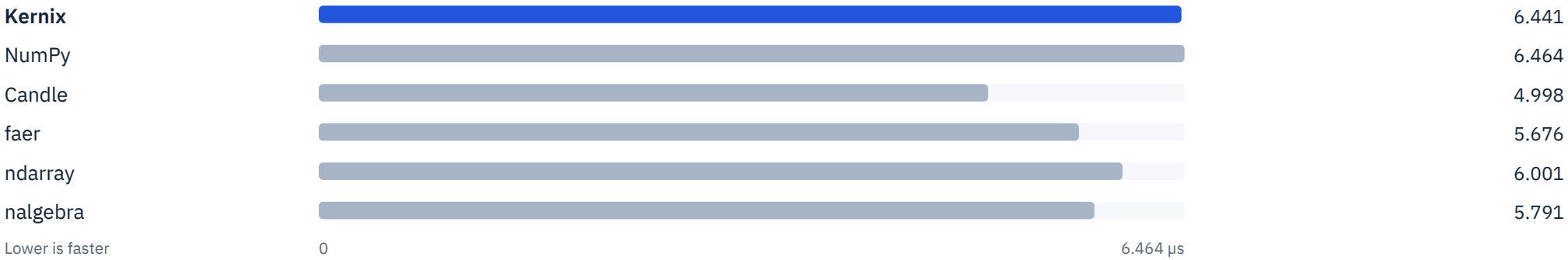
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



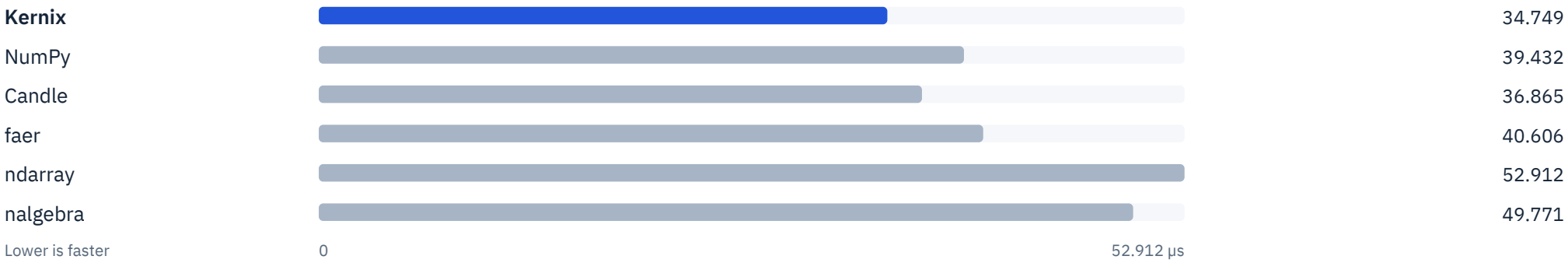
Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

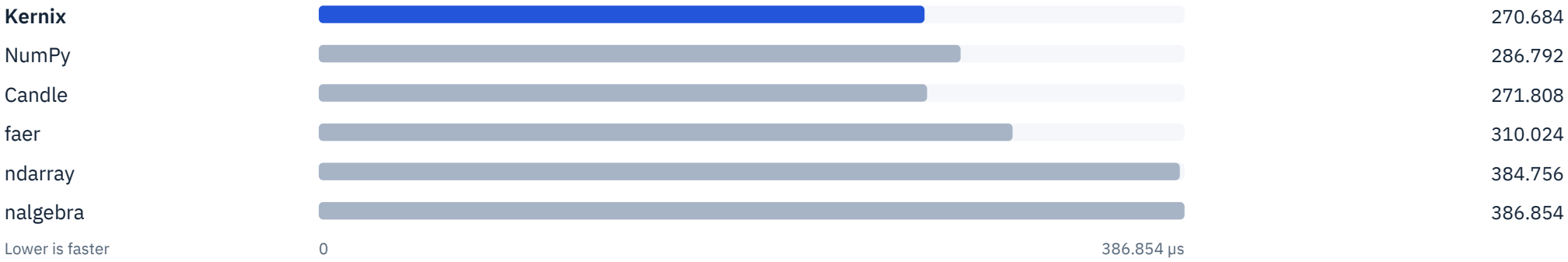
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

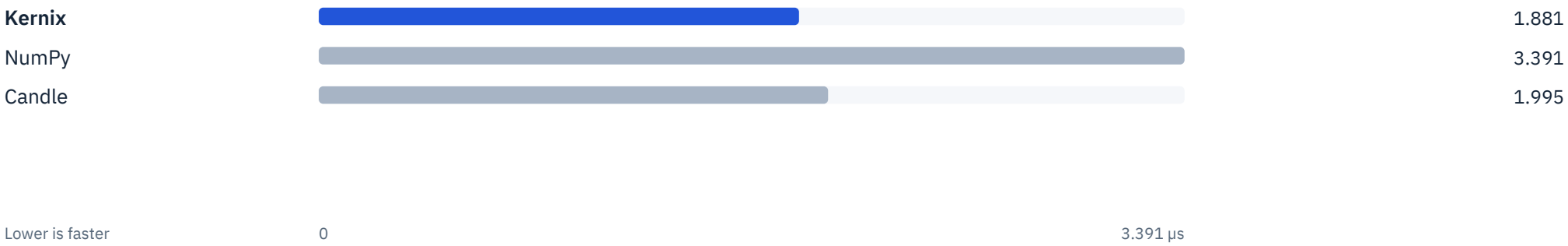


Kernix 0.7.3 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



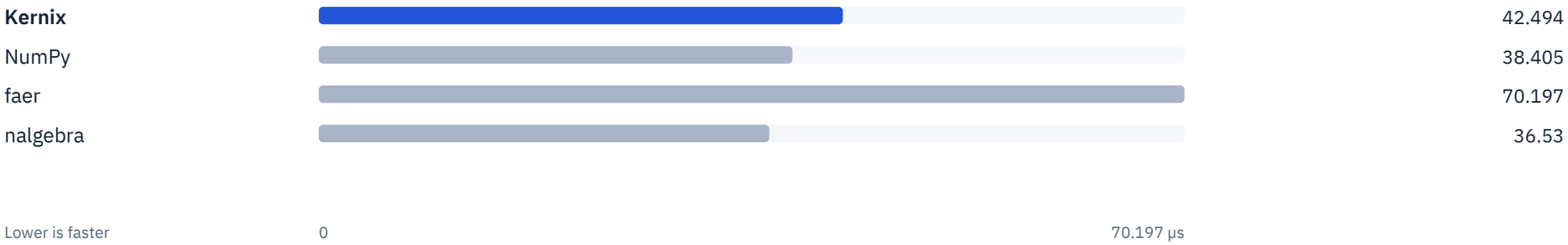
Kernix 0.7.3 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

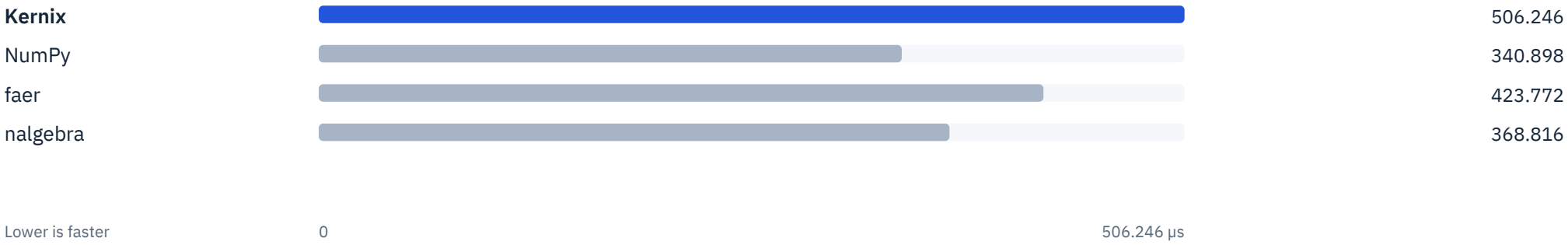
Per-call median



Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median

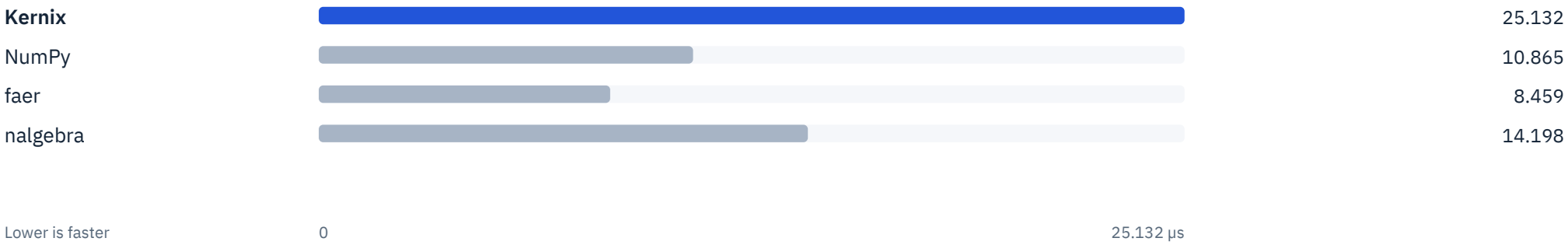


Kernix 0.7.3 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

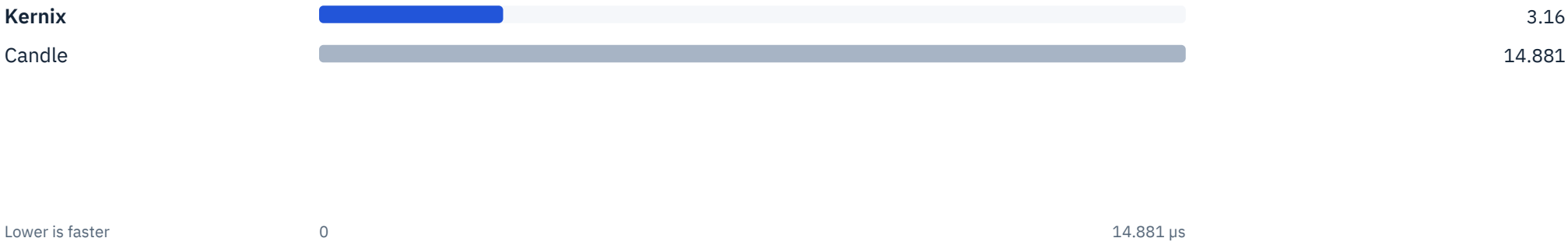
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.3 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

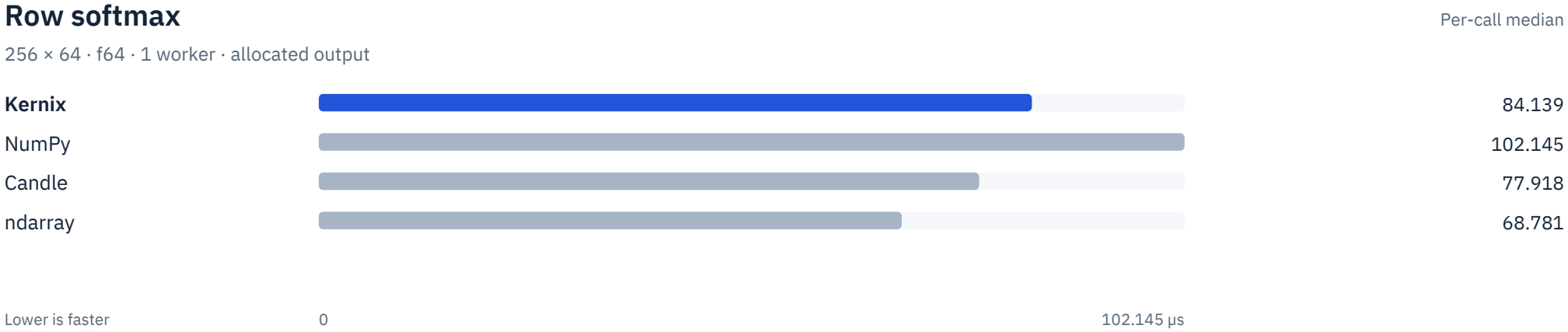


Kernix 0.7.3 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

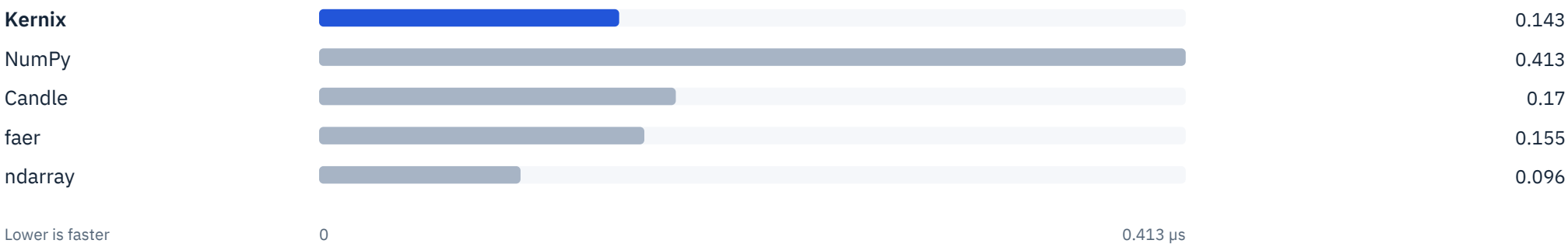


Kernix 0.7.3 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1535	1.535
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1492	1.492
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1612	1.612
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1600	1.6
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1525	1.525
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2808	2.808
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9162	9.162
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	11312	11.312
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11075	11.075
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10701	10.701

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9671	9.671
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12827	12.827
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	73624	73.624
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	117855	117.855
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115569	115.569
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	82494	82.494
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	73614	73.614
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	78457	78.457
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	574036	574.036
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	665964	665.964

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	662924	662.924
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	722614	722.614
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	575843	575.843
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	612150	612.15
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6441	6.441
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6001	6.001
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	5791	5.791
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5676	5.676
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4998	4.998
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6464	6.464

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34749	34.749
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	52912	52.912
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49771	49.771
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	40606	40.606
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	36865	36.865
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39432	39.432
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	270684	270.684
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	384756	384.756
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	386854	386.854
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	310024	310.024

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	271808	271.808
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	286792	286.792
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1881	1.881
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1995	1.995
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3391	3.391
Thin QR / Kernix 96 × 32 · f64	42494	42.494
Thin QR / nalgebra 96 × 32 · f64	36530	36.53
Thin QR / faer 96 × 32 · f64	70197	70.197
Thin QR / NumPy 96 × 32 · f64	38405	38.405
Thin QR / Kernix 256 × 64 · f64	506246	506.246

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / nalgebra 256 × 64 · f64	368816	368.816
Thin QR / faer 256 × 64 · f64	423772	423.772
Thin QR / NumPy 256 × 64 · f64	340898	340.898
Cholesky factorization / Kernix 64 × 64 · f64	25132	25.132
Cholesky factorization / nalgebra 64 × 64 · f64	14198	14.198
Cholesky factorization / faer 64 × 64 · f64	8459	8.459
Cholesky factorization / NumPy 64 × 64 · f64	10865	10.865
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2378	2.378
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	4273	4.273
Model inference / Kernix batched normalized/sliced MLP · f64	2029	2.029

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	3582	3.582
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2699	2.699
Model recording + forward / Candle batched normalized/sliced MLP · f64	5252	5.252
Model backward / Kernix prepared graph → fresh gradients · f64	3160	3.16
Model backward / Candle prepared graph → fresh gradients · f64	14881	14.881
Parameter update / Kernix transactional SGD, fixed gradients · f64	528	0.528
Row softmax / Kernix 256 × 64 · f64	84139	84.139
Row softmax / ndarray 256 × 64 · f64	68781	68.781
Row softmax / Candle 256 × 64 · f64	77918	77.918
Row softmax / NumPy 256 × 64 · f64	102145	102.145

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	143	0.143
Create zeros / ndarray 64 × 64 · f32	96	0.096
Create zeros / faer 64 × 64 · f32	155	0.155
Create zeros / Candle 64 × 64 · f32	170	0.17
Create zeros / NumPy 64 × 64 · f32	413	0.413
Create ones / Kernix 64 × 64 · f32	185	0.185
Create ones / ndarray 64 × 64 · f32	144	0.144
Create ones / faer 64 × 64 · f32	181	0.181
Create ones / Candle 64 × 64 · f32	231	0.231
Create ones / NumPy 64 × 64 · f32	1390	1.39